Project Title

Credit Card Transaction Fraud Detection

Data Set

The Credit Card Fraud Detection Dataset (2023 / 2025 Variants) available on Kaggle. These datasets feature transaction metrics, such as transaction time, localized timestamp markers, physical/digital merchant categories, dollar amounts, and location identifiers, with a naturally heavy class imbalance (~1% to 3% fraud distribution).

Project Idea

Most conventional credit card fraud detection tutorials treat transaction logs as static spreadsheets, evaluating each purchase in isolation using standard ML models. In real-world financial systems, however, identifying fraud relies heavily on dynamic context—such as transaction **velocity** (e.g., number of swipes within a 10-minute window) and **spatiotemporal physics** (e.g., verifying if physical distance jumps between consecutive merchant coordinates represent impossible travel).

To address this, this project implements a **Stateful Streaming Feature Simulator** in Python that processes sorted transaction histories in real time, maintaining sliding state windows per cardholder to derive dynamic velocity and Haversine distance metrics. To capture both temporal sequence dynamics and complex tabular feature boundaries under extreme class imbalance (~1% to 3% fraud ratio), the architecture leverages a hybrid modeling approach—combining a **Long Short-Term Memory (LSTM)** network to learn sequential behavioral representations with a **Balanced Random Forest** (or LightGBM) classifier.

Finally, predictions are passed through a **Decoupled Authorization Rule Engine** that merges machine learning risk probabilities with hard business rules (e.g., instant velocity limits and high-risk geo-fencing) to simulate a live, low-latency transaction approval pipeline before charges settle.

Software You Will Need to Write

1. **Stateful Streaming Feature Simulator:** A Python script utilizing pandas and numpy that iterates through sorted transaction logs, maintaining an in-memory sliding window state per user to generate real-time velocity metrics (e.g., swipe frequency) and spatiotemporal physics metrics (e.g., Haversine distance and travel speed between consecutive merchant coordinates).

2. **Sequential LSTM Embedding Extractor:** A deep learning module built with PyTorch or TensorFlow that processes chronological sliding sequence buffers of past N transactions per cardholder to learn dense latent embeddings representing shifting temporal user behavior.

3. **Hybrid Imbalanced Training & Fusion Pipeline:** A machine learning pipeline using scikit-learn and imbalanced-learn that combines the generated LSTM temporal embeddings with static tabular features, training a **Balanced Random Forest** (or LightGBM) classifier to optimize precision-recall trade-offs under extreme class imbalance (~1% to 3% fraud ratio).

4. **Decoupled Authorization Rule Engine:** An evaluation script that blends the hybrid model's predicted fraud risk probabilities with hard business rules (e.g., auto-declining transactions exceeding impossible physical travel speed thresholds) to output real-time, low-latency approve/challenge/decline decisions.

Papers to Read

1. *Hybrid LSTM and Ensemble Model for Credit Card Fraud Detection*

- **Core Architecture:** Fuses a sequence-aware **LSTM with Attention** and a **Calibrated Random Forest** into a weighted ensemble score:

$$P_{\text{final}} = w_1 \cdot P_{\text{LSTM}} + w_2 \cdot P_{\text{RF}}$$

2. *Credit Card Fraud Detection Using a Stacking Ensemble Approach with LSTM and Random Forest*

- **Core Architecture:** A **Stacking Ensemble** where **LSTM** and **Random Forest** act as base learners, and a Multi-Layer Perceptron (MLP) acts as the meta-classifier.

3. *Credit Card Fraud Detection with Autoencoder and Probabilistic Random Forest (AE-PRF)*

- **Core Architecture: Deep Feature Extractor + Probabilistic Random Forest**.

4. *Sequence-to-Tabular Ensembling for High-Frequency Transaction Fraud*

- **Core Architecture:** Dual-branch framework evaluating sequential sliding windows vs. point-in-time streaming metrics.

Teammate

Han Tun Zaw (615915), Pabin Luitel (618977)

What Will You Complete by __?

By the milestone date, I will complete the implementation of the **Stateful Streaming Feature Simulator** to calculate real-time transaction velocity and Haversine physical distance metrics from sorted transaction logs. I will also construct the chronological sequence window buffering pipeline and train the initial **LSTM network** to generate latent temporal user embeddings.

Additionally, I will share my preliminary experimental evaluation, presenting a performance benchmark (using Precision-Recall AUC and $F1$-score) that compares a standard baseline classifier trained on raw features against the hybrid **LSTM + Balanced Random Forest** architecture operating on engineered streaming and sequence representations.

References

● Kaggle: Credit Card Fraud Detection Dataset documentation.